

Waiz Khan

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EDUCATION

Johns Hopkins University

M.S.E. in Data Science

- Coursework: Machine Translation, Artificial Intelligence, Deep Learning, Statistical Learning

Baltimore, MD

Aug. 2024 – May 2027

GD Goenka University

B.Tech. in Computer Science & Engineering

Gurugram, India

Aug. 2019 – Nov. 2023

RESEARCH EXPERIENCE

Research Assistant

Johns Hopkins University, Advisor: Prof. Philipp Koehn

- Design and validate evaluation metrics for LLM representations: built an intrinsic multilinguality metric and tested it across 17 open models against 30+ preregistered predictions and held out model families
- Run the experimental program on a Slurm GPU cluster, from large scale representation extraction to intervention studies across 6 training objectives and 3 seeds
- Author the technical reports behind an NSF proposal currently in development with Prof. Koehn
- Earlier phase: benchmarked 15+ decoder architectures and learning rules for hieroglyph to text translation on a 61k example corpus; best decoders reach BLEU-4 30.3, ten times the Transformer baseline

Dec. 2025 – Present

Baltimore, MD

Research Assistant

GD Goenka University, Advisor: Dr. Yogesh Kumar

- Studied chronic school absenteeism on the IHDS-II national survey of 40k+ students; logistic regression, KNN and random forest models reached 74% accuracy under fair validation

Aug. 2022 – Nov. 2022

Gurugram, India

Data Scientist Intern

EBMS Solutions Pvt Ltd

- Engineered a churn prediction pipeline (XGBoost, time series cross validation) deployed to a production CRM with automated retraining; contributed to a +15% retention lift at AUC 0.82

May 2022 – Sep. 2022

Chennai, India

PROJECTS

AttendOpt | *Python, CatBoost, DoWhy, PuLP*

Jan. 2026

- Extension of my 2022 absenteeism research into an end to end pipeline on IHDS-II data (47,027 students): an Optuna tuned gradient boosting ensemble reaches ROC-AUC 0.841 against a 0.699 logistic baseline, calibrated with Platt and isotonic scaling
- Validated the distance barrier causally with DoWhy (ATE 0.14 per SD, refutation tests pass); an ILP allocator doubles expected impact against random allocation at a fixed budget

ExoSAGE | *Python, FastAPI, D3*

Jul. 2026

- Point in time screening of NASA TOI transit candidates with false discovery rate control and walk forward backtesting, plus mixed integer telescope hour allocation under correlated risk
- 11 test suites over checksummed data snapshots and real TESS light curves; reported as an observability screen, not a trained classifier

Black-Scholes Pricer | *Python, NumPy, SciPy, Streamlit*

Jan. 2025

- European options pricer with Greeks, a Newton-Raphson implied volatility solver with Brent fallback, and arbitrage diagnostics; 91+ tests in CI

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, C++, Swift, Bash

LLMs & Agents: agent pipelines and tool use, RAG and embedding retrieval, prompt engineering, agent evaluation harnesses and safety gating, Hugging Face Transformers

ML: PyTorch, scikit-learn, XGBoost, CatBoost; transformers, CNNs; causal inference (DoWhy); experimental design, preregistered evaluation, time series cross validation

Systems & Data: Node.js, Next.js, SQLite/PostgreSQL, pandas/NumPy/SciPy, Slurm HPC clusters, Git, Docker